National University
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Foundation of Advancement
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Lab Number:	09
Course Code:	CL2001
Course Title:	Data Structures - Lab
Semester:	Fall 2026
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Guidelines

- You may use **only the concepts covered in the lab** to complete this task.
- Your submission must be a **single PDF file** that includes:
  - The source code (in plain text)
  - Screenshots of the corresponding outputs
- Name your file using the following format: **rollno_name_labtasknumber.pdf**
  Example: *25p-1234_ozair_labtask09.pdf*
- Unethical use of AI tools will result in **0 marks**.
- Late submissions will incur a penalty of **-3 marks per day**.

Tasks

NOTE: USE OF GLOBAL OR STATIC VARIABLES OR EXTRA FUNCTION PARAMETERS IS NOT ALLOWED

1 Basic BST Operations

Create a Binary Search Tree (BST) to store integers. Insert the values `50, 30, 70, 20, 40, 60` into the tree in the given order.

Display the values stored in the tree using preorder, inorder, postorder and level-order traversals.

Example:

```
Input:
            50
           /  \
         30    70
        / \    /
       20 40  60

Output:
Preorder:   50 30 20 40 70 60
Inorder:    20 30 40 50 60 70
Postorder:  20 40 30 60 70 50
Levelorder: 50 30 70 20 40 60
```

2 Find Minimum and Maximum

Write functions to find and return the minimum and maximum values stored in a BST. Your solution should use the BST property to locate these values without traversing the entire tree.

Example:

```
Minimum = 20
Maximum = 70
```

3 Search for an Element

Given a BST containing integers, write a function that searches for a specified value and returns `true` if the value exists in the tree or `false` otherwise.

Example 1:

```
Input: SearchItem = 40
Output: true
```

Example 2:

```
Input: SearchItem = 90
Output: false
```

4 Count Nodes

Write recursive functions to count the total number of nodes and the number of leaf nodes in a BST. A leaf node is a node that has no children.

Example:

```
Total Nodes = 6
Leaf Nodes = 3
```

5 Calculate Tree Height

Write a recursive function that calculates and returns the height of a BST. For this task, the height of an empty tree is -1, and the height of a tree containing only one node is 0.

Example 1:

```
Given Root: 50
```

```
Height = 2
```

Example 2:

```
Given Root: 30
Height = 1
```

6 Validate a BST

Write a function that determines whether a given binary tree satisfies the Binary Search Tree (BST) property.

For every node, all values in its left subtree must be smaller than the node's value, and all values in its right subtree must be greater. This property must hold throughout the entire tree, not just between a node and its immediate children.

Example 1:

```
      10
     /  \
    5    15

Output: true
```

Example 2:

```
     10
    /  \
   5    15
       /
      6

Output: false
```

7 Find the Kth Smallest Element

Given a BST containing distinct integers and an integer k, write a function that returns the kth smallest element in the tree. Assume that k is a valid position in the ascending order of the values stored in the tree.

Example:

```
Input: k = 3
Output: Kth Smallest = 40
```

8 Find Values Within a Range

Given a BST and two integers, `low` and `high`, display all values within the inclusive range **[low, high]** in ascending order.

Your solution should use the BST property to avoid visiting subtrees that cannot contain values within the specified range.

Example:

```
Input: low = 35, high = 65
Output: Values in Range: 40 50 60
```